CHAPTER 20 · DERIVATIONS

OSCILLATIONS

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 VELOCITY IN TERMS OF DISPLACEMENT

SYLLABUS 20.1(C)

RESULT $v = \pm\omega\sqrt{A^2 - x^2}$

1. **Method 1 (from $a = -\omega^2 x$)**: write $a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$, so $v\frac{dv}{dx} = -\omega^2 x$.
2. Separate and integrate: $\int v\,dv = -\omega^2 \int x\,dx$, using $v = 0$ when $x = A$ (extreme position).
3. $\frac{1}{2}v^2 = \frac{1}{2}\omega^2(A^2 - x^2)$, so $\mathbf{v = \pm\omega\sqrt{A^2 - x^2}}$.
4. **Method 2 (from $x = A\sin\omega t$)**: $\sin\omega t = \frac{x}{A}$ and $v = \frac{dx}{dt} = A\omega\cos\omega t$, so $\cos\omega t = \frac{v}{A\omega}$.
5. Use $\sin^2\omega t + \cos^2\omega t = 1$: $\frac{x^2}{A^2} + \frac{v^2}{A^2\omega^2} = 1$, which again gives $\mathbf{v = \pm\omega\sqrt{A^2 - x^2}}$; maximum ωA at $x = 0$, zero at $x = \pm A$.

2 ACCELERATION IN TERMS OF TIME, AND SHOWING THE MOTION IS SHM

SYLLABUS 20.2

RESULT $a = -\omega^2 A\sin\omega t = -\omega^2 x$

1. Given the displacement $x = A\sin\omega t$.
2. Differentiate x with respect to t: $v = \frac{dx}{dt} = A\omega\cos\omega t$.
3. Differentiate v with respect to t: $\mathbf{a = \frac{dv}{dt} = -A\omega^2\sin\omega t}$.
4. Since $A\sin\omega t = x$, $\mathbf{a = -\omega^2 x}$: the acceleration is directly proportional to the displacement and always directed towards the equilibrium position, so the motion is SHM.

3 KINETIC AND POTENTIAL ENERGY IN SHM

SYLLABUS 20.2(A)

RESULT $K = \frac{1}{2}m\omega^2(A^2 - x^2)$, $U = \frac{1}{2}m\omega^2 x^2$

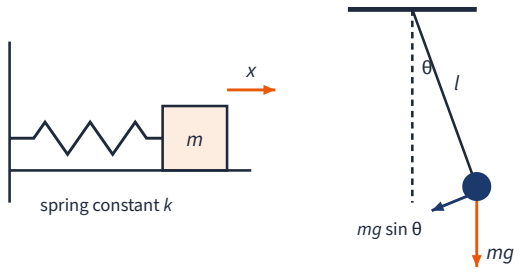
1. Use $K = \frac{1}{2}mv^2$ with $v^2 = \omega^2(A^2 - x^2)$: $\mathbf{K = \frac{1}{2}m\omega^2(A^2 - x^2)}$.
2. K is maximum at $x = 0$, where $U = 0$, so the total energy is $E = K_{max} = \frac{1}{2}m\omega^2 A^2$.
3. Principle of conservation of energy at any position: $U = E - K = \frac{1}{2}m\omega^2 A^2 - \frac{1}{2}m\omega^2(A^2 - x^2) = \mathbf{\frac{1}{2}m\omega^2 x^2}$.
4. In terms of t ($x = A\sin\omega t$): $K = \frac{1}{2}m\omega^2 A^2\cos^2\omega t$ and $U = \frac{1}{2}m\omega^2 A^2\sin^2\omega t$; $K + U = \frac{1}{2}m\omega^2 A^2$ is constant.

4 SHOWING A SYSTEM PERFORMS SHM: SPRING-MASS SYSTEM AND SIMPLE PENDULUM

SYLLABUS 20.3(A)

RESULT $T = 2\pi\sqrt{\frac{m}{k}}$, $T = 2\pi\sqrt{\frac{l}{g}}$

1. **Method**: write the restoring (or resultant) force on the body at displacement x, put it equal to ma, and show that a = −(constant)x. Then ω^2 = constant and $T = \frac{2\pi}{\omega}$.
2. **Spring–mass**: displaced x from equilibrium, restoring force F = −kx. ma = −kx, so $a = -\frac{k}{m}x$.
3. This is SHM ($a = -\omega^2 x$) with $\omega^2 = \frac{k}{m}$: $\mathbf{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}}$.
4. **Simple pendulum**: displaced by angle θ, restoring force along the arc = −mg sin θ ≈ −mgθ for small θ, and $\theta = \frac{x}{l}$.
5. $ma = -\frac{mgx}{l}$, so $a = -\frac{g}{l}x$: $\omega^2 = \frac{g}{l}$ and $\mathbf{T = 2\pi\sqrt{\frac{l}{g}}}$.



Valid for small oscillations (θ less than about 10°).